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POME Unit - 5

📊 Macroeconomic Indicators

1️⃣ Prices and Inflation

1.1 🏷️ Price & Price Level

🔹 What are “Prices” in Macroeconomics?

In macroeconomics, **prices** refer to the **money value assigned to goods and services**.

They are primarily determined by:

> **Supply and Demand**

- 📈 **High Demand + Low Supply → Price Increases**
- 📉 **Low Demand + High Supply → Price Decreases**

🔎 Why Prices Matter?

Prices act as **signals** in the economy.

They:

- Guide **what to produce**
- Influence **consumer decisions**
- Affect **resource allocation**

📌 Example:

If **crude oil prices rise**, people:

- Seek alternatives (electric vehicles, public transport)
- Reduce consumption
- Businesses adjust production strategies

🔹 What is the Price Level?

The **price level** is:

> The **average cost of goods and services** in an economy.

It provides:

- A broad measure of **purchasing power**
- Insight into **inflation trends**

📏 Tools Used to Measure Price Level

Tool	What It Measures	Scope
Consumer Price Index (CPI)	Price changes in a fixed basket of everyday goods	Consumer-level
GDP Deflator	Price changes of *all* goods and services produced	Economy-wide

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1.2 ⏳ Price Level and Time

The **price level changes over time** due to economic conditions.

📜 Historical Examples

Event	What Happened	Impact on Price Level
Great Depression	Demand collapsed	Prices fell (Deflation)
Oil Crisis	Oil shortages	Production costs rose → Inflation surged
COVID-19	Supply chain disruptions	Prices increased due to shortages

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⚙️ Important Insight

Not all prices continuously rise.

Technological innovation can **reduce costs**, even if overall price levels increase.

📱 Example:

- Electronics have become cheaper
- Better efficiency + mass production
- Improved features at lower prices

🎯 Why Understanding Price Level Changes Matters

It helps manage:

- Inflation
- Deflation
- Economic growth stability

1.3 📊 Price Index

🔹 Definition

A **price index** measures:

> How the average cost of a fixed set of goods changes over time.

It is used to track:

- Inflation
- Deflation trends

🏛️ Policy Relevance

Governments use price indices to adjust:

- Taxes
- Interest rates
- Government spending

Purpose:

> Stabilize the economy

1.4 🛒 Consumer Price Index (CPI)

🔹 Definition

The **Consumer Price Index (CPI)** is a key tool for measuring inflation.

It tracks price changes in a **fixed basket of everyday goods and services**, including:

- 🍞 Food
- 👕 Clothing
- 🏠 Housing
- 🏥 Healthcare
- 🚗 Transportation

🔹 Why CPI is Important

CPI is used to adjust:

- Wages
- Pensions
- Government payments

So that they:

> Keep up with the cost of living

1.5 ⚠️ Problems with CPI

Although useful, CPI has limitations.

🔹 Main Issue: Quality Change Bias

📱 Example:

- Smartphones evolve rapidly.
- CPI may show stable prices.
- But new phones have **better features**.

If CPI does not account for quality improvements:

- It may **overestimate inflation**
- It may suggest prices are rising
- When consumers are actually getting better value

🔎 Core Limitation

CPI may not fully capture:

> The true change in the cost of living

1.6 📈 Inflation

🔹 Definition

Inflation is:

> The percentage increase in the price index over time.

It represents:

- A decline in purchasing power
- Money buying fewer goods than before

📏 Measurement

Inflation is typically measured annually using:

- CPI
- Other price indices

🎯 Why Inflation Management Matters

It is crucial for:

- Economic stability
- Sustainable growth

Managed using:

- **Monetary Policy**
- **Fiscal Policy**

2️⃣ Exchange Rate

2.1 💱 Definition

The **exchange rate** is:

> The price of one currency in terms of another.

It plays a crucial role in:

- International trade
- Economic relations between countries

2.2 🌍 Exchange Rate Systems

A country's exchange rate system determines how its currency value is managed.

🅐 Flexible Exchange Rate System

📌 Definition:

- Determined by supply and demand in the market.

📍 Example:

- United States (US Dollar)

✅ Pros:

- Adapts to market conditions

🅑 Fixed Exchange Rate System

📌 Definition:

- Government or central bank pegs currency to another.

📍 Example:

- Hong Kong (HKD fixed at 7.8 USD)

✅ Pros:

- Stability benefits trade and investment

🅒 Managed Float System

📌 Definition:

- Exchange rates fluctuate
- Central bank intervenes when necessary

📍 Example:

- India

Central bank steps in during sharp fluctuations.

✅ Pros:

- Balances flexibility and stability

❌ Cons:

- Requires constant monitoring

🅓 Monetary Union

📌 Definition:

- Multiple countries use a single currency.
- Eliminates internal exchange rates.

📍 Example:

- European Union (Euro €)

✅ Pros:

- No exchange rate risks within the union

❌ Cons:

- Countries lose individual control over monetary policy

2.5 🌐 Effective Exchange Rate

🔹 Definition

The **effective exchange rate** measures:

> A currency's strength against multiple currencies, not just one.

It gives a broader view of economic performance.

🔎 Interpretation (Example: Rupee)

Situation	Meaning	Impact
Effective Exchange Rate ↑	Rupee Appreciates	Indian exports become expensive
Effective Exchange Rate ↓	Rupee Depreciates	Indian goods cheaper abroad → Trade boosts

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3️⃣ Gross Domestic Product (GDP)

3.1 📦 Definition

Gross Domestic Product (GDP) is:

> The total value of all final goods and services produced within a country during a specific period (usually a year or quarter).

It reflects:

- Overall economic activity
- Total output

🔹 Important Rule: Final Goods Only

GDP includes:

- Only **final goods and services**
- To avoid **double counting**

🚗 Example of Avoiding Double Counting

If:

- A car manufacturer buys tires from a supplier

Then:

- Value of tires is **not counted separately**
- Only the final price of the car is included in GDP

📦 4. The Components of GDP

4.1 🔄 The Circular Flow – Simple Version

📌 Core Idea

Gross Domestic Product (GDP) represents:

> The total market value of all final goods and services produced within a country during a specific period.

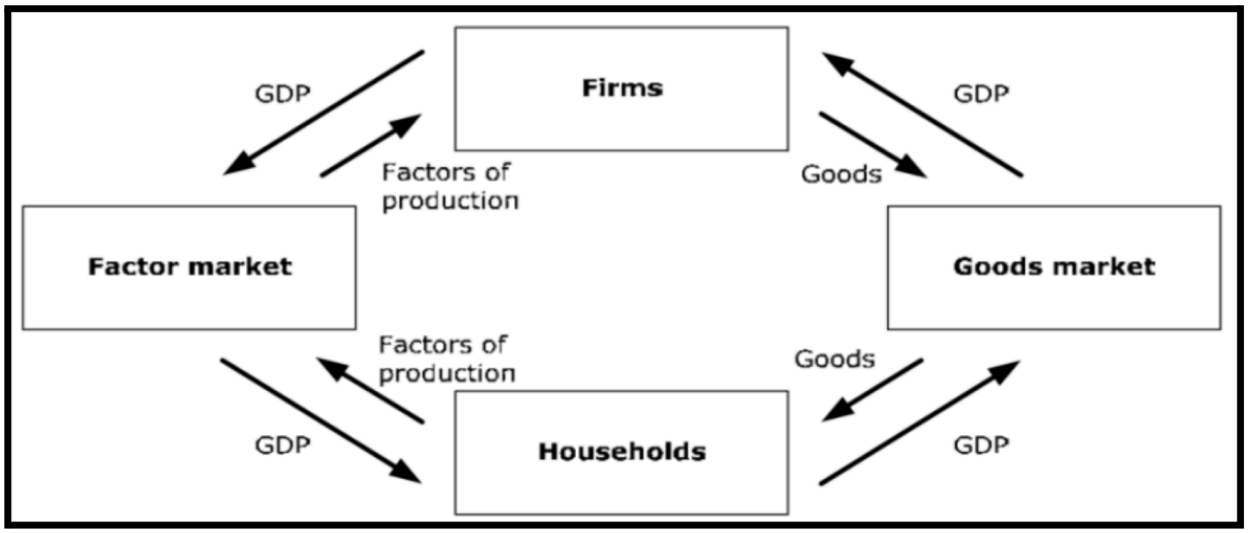
To understand how GDP functions in the economy, we use the **Circular Flow Model**.

🔁 Direction of Flows

Flow Type	Direction
Goods & Services	Clockwise
Money	Counterclockwise

This opposite movement ensures continuous economic activity.

🏭 How the Circular Flow Works



Step-by-Step Movement

1️⃣ **Firms produce finished goods**

→ Send them to the **Goods Market**

2️⃣ **Consumers purchase goods**

→ Money flows to firms

→ This money equals **GDP**

3️⃣ Consumers provide:

- Labor
- Capital
- Other factors of production

→ Supplied through **Factor Markets**

4️⃣ Firms pay:

- Wages
- Rent
- Interest
- Profits

→ Using income earned from selling goods

🎯 Why It Matters

This continuous movement of:

- Resources
- Goods
- Money

Keeps the economy:

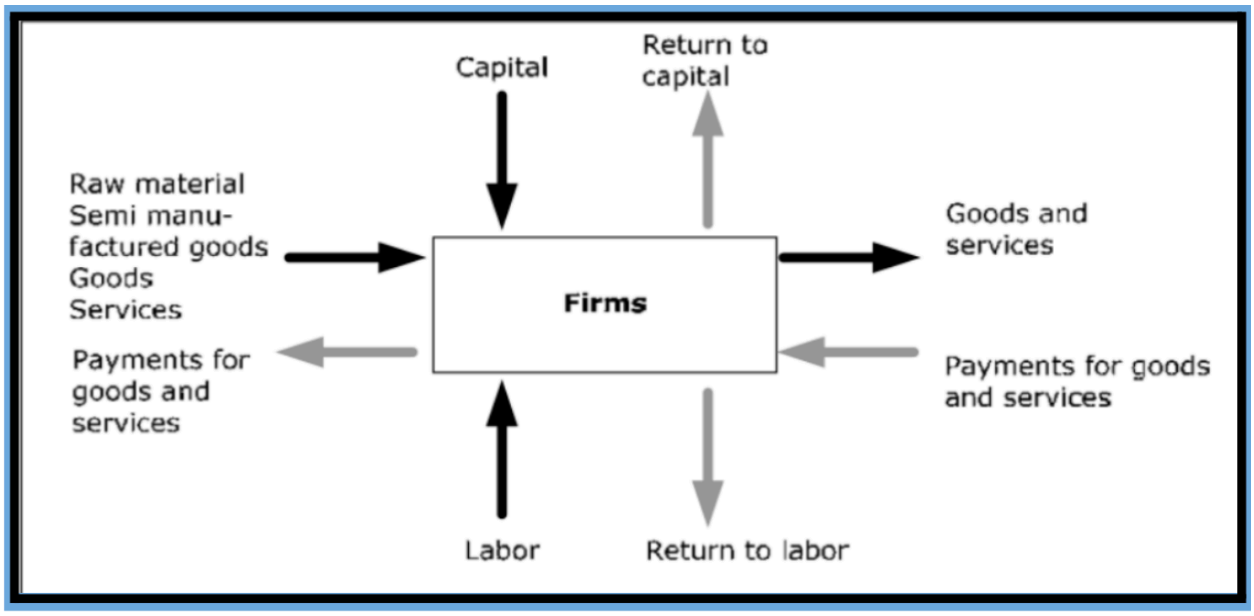
- Running
- Stable
- Productive

4.2 🏭 Firms and the Concept of Value Added

📌 Core Principle

To avoid **double counting**, GDP counts:

> Only the **value added at each stage of production**



🧁 Example 1: A Bakery

Let's trace the production of bread.

Stage 1: Raw Inputs

- Bakery buys flour, yeast, water
- This is the **first value-added stage**

Stage 2: Processing

- Pays workers
- Pays rent
- Converts ingredients into dough
- Bakes bread

→ More value is added

Stage 3: Final Sale

- Bread sold to consumers
- Price reflects **total value added**

🚫 Why Not Count Everything?

If we counted:

- Flour value
- Dough value
- Bread value

We would inflate GDP.

Instead:

> GDP = Final Value Added Only

4.3 🔄 Circular Flow – Circulation of Firms

Firms are grouped into **three production stages**:

Type	Meaning	Role
FR	Firms acquiring raw materials	Extract raw materials
FH	Firms producing semi-manufactured goods	Process raw materials
FF	Firms producing finished goods	Create final products

📊 GDP Flow Through Production Stages

Let total GDP = **Y**

Stage	Value Expression	Meaning
From FR → FH	**YR**	Value of raw materials
Value added by FH	**YH − YR**	Semi-manufacturing stage
Value added by FF	**Y − YH**	Final production stage

🧠 Key Insight

If we sum:

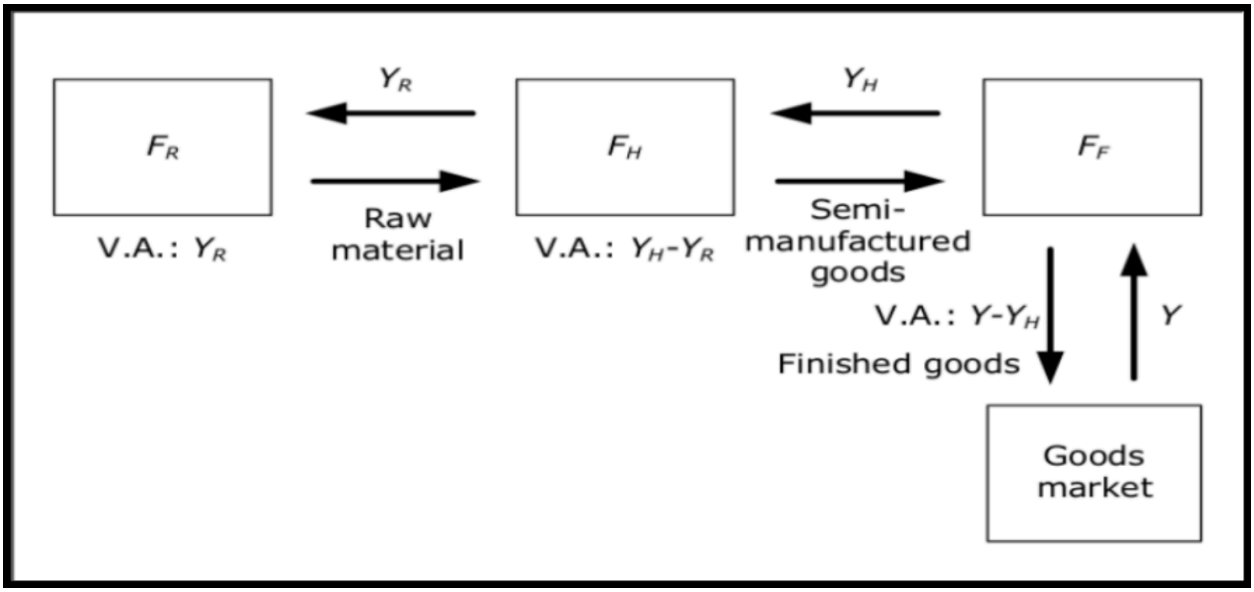
- YR

- (YH − YR)
- (Y − YH)

We get:

> Total GDP (Y)

This ensures accurate measurement without duplication.



📱 Example: Smartphone Manufacturing

1️⃣ Raw materials

- Metals
- Glass
- Microchips

2️⃣ Semi-processing

- Component assembly
- Factory rent
- Worker wages

- Utilities

3️⃣ Final Product

- Finished smartphone sold to consumers

Money flows back to:

- Firm
- Workers
- Suppliers

And production continues.

4.4 📦 Circular Flow – Circulation of Goods

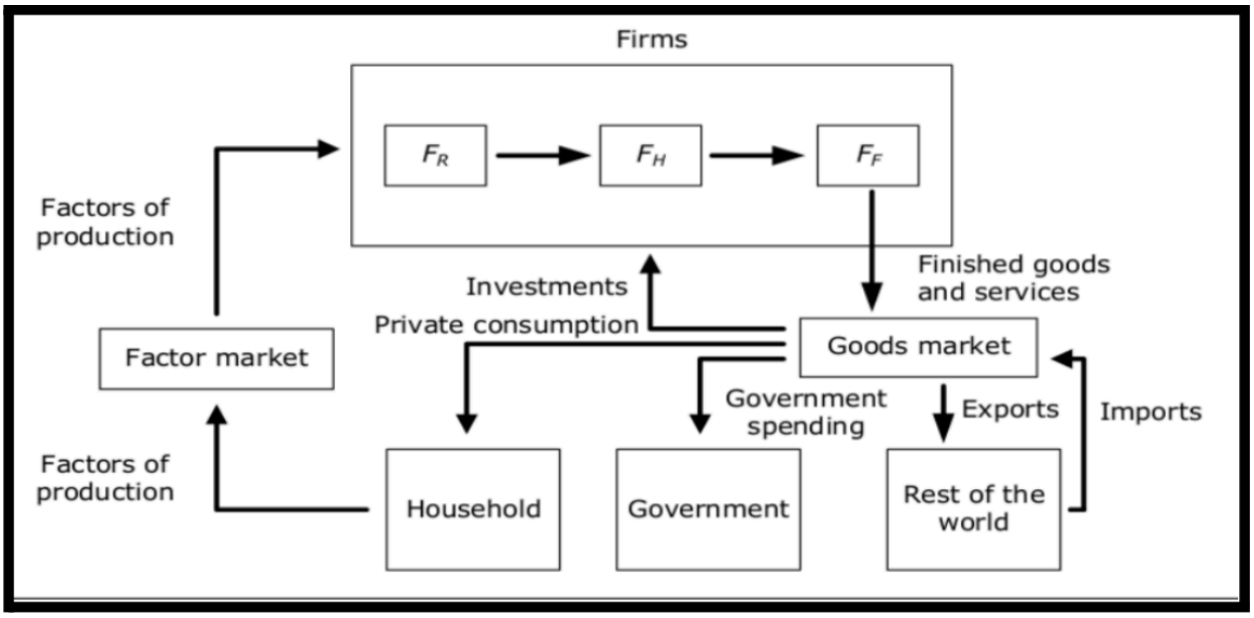
Finished goods are categorized into:

Category	Description	Sector
Private Consumption (C)	Goods bought by households	Household sector
Public Consumption (G)	Goods bought by government	Government sector
Investment (I)	Business expansion spending	Business sector
Exports (X)	Goods sold abroad	Foreign sector
Imports (Im)	Goods bought from abroad	Foreign sector

📌 Total Circular Flow of Goods

$$\text{Total Goods Flow} = C + G + I + X - Im$$

This shows how different sectors contribute to GDP.



4.5 💰 Circular Flow – Circulation of Money

Money circulates in the opposite direction of goods.

Flow of Money

Firms pay:

- Wages
- Rent
- Interest
- Profits

Households use income to:

- Buy goods and services

Government collects:

- Taxes
- Spends on public goods

Foreign sector affects:

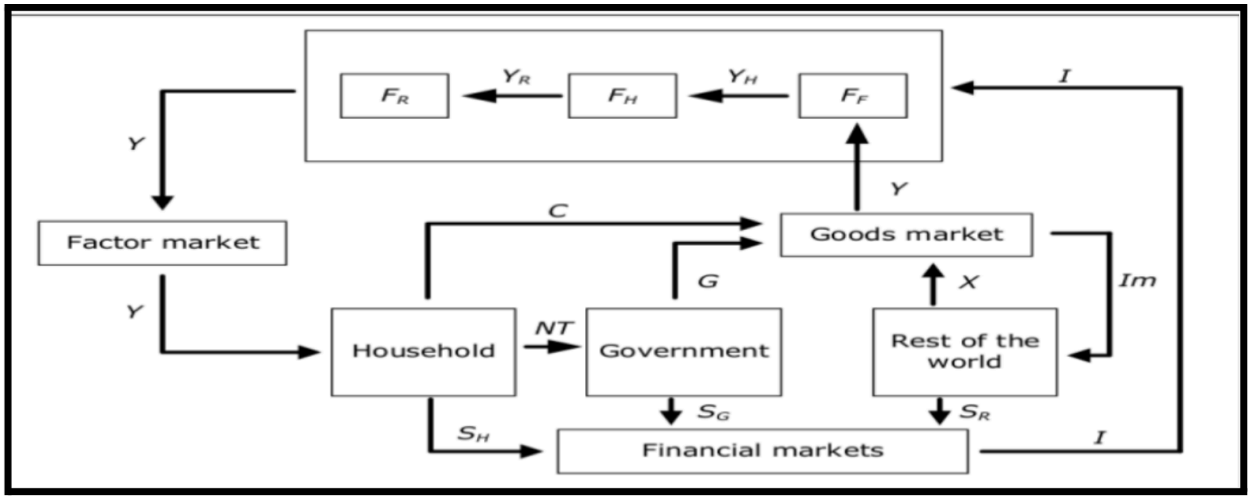
- Exports
- Imports

📌 Total Circular Flow of Money

$$\text{Total Money Flow} = HouseholdIncome + FirmExpenditure + GovernmentRevenue + NetExports$$

This supports:

- Production
- Consumption
- Economic growth



4.6 📊 Components of GDP

1️⃣ Total Output Identity

$$Y + Im = C + I + G + X$$

Where:

Symbol	Meaning
Y	GDP (Total Output)
Im	Imports
C	Consumption
I	Investment
G	Government Spending
X	Exports

🧠 Interpretation

Left Side:

- All finished goods entering the economy

Right Side:

- Their four uses:
  - Consumption
  - Investment
  - Government Spending
  - Exports

2️⃣ Net Exports (NX)

Rearranging:

$$Y = C + I + G + X - Im$$

Define:

$$NX = X - Im$$

So:

$$Y = C + I + G + NX$$

🔎 What Net Exports Mean

Condition	Interpretation
NX > 0	Trade Surplus
NX < 0	Trade Deficit

It shows whether a country:

- Sells more abroad
- Or buys more from abroad

3️⃣ Final GDP Identity

$$\boxed{Y = C + I + G + NX}$$

The Four Major Components

Component	Meaning
C	Household consumption
I	Business investment
G	Government spending
NX	Net exports

4.7 📏 Four Different Measures of GDP (EPIC)

The circular flow model allows GDP to be measured in **four ways**.

1️⃣ Expenditure (Spending) Method

GDP = Total spending on final goods and services

$$Y = C + I + G + X - Im$$

Includes:

- Consumption
- Investment
- Government Spending
- Net Exports

2️⃣ Production (Value Added Method)

GDP = Sum of value added at each production stage

✔️ Prevents double counting

✔️ Measures sector contribution

3️⃣ Components (Summation) Method

Direct breakdown:

$$Y = C + I + G + X - Im$$

Focuses on structural decomposition.

4️⃣ Income (Earnings) Method

GDP = Total income earned in production

$$Y = W + R + I + P$$

Where:

Symbol	Meaning
W	Wages
R	Rent
I	Interest
P	Profits

🎯 What This Method Shows

It highlights:

- Income distribution
- Who earns from production
- How output translates into income

🧠 Master Summary Table

Perspective	Formula	Focus
Expenditure	C + I + G + NX	Spending
Production	Value Added	Output
Components	C + I + G + NX	Structural Breakdown
Income	W + R + I + P	Earnings

📘 4.9 Capital

📌 What is Capital?

In economics, **capital** refers to:

> Manufactured goods used to produce other goods and services
> (but not consumed during production)

Examples:

- 🏭 Machinery
- 💻 Computers
- 🏢 Factories
- 🛣️ Infrastructure

⚠️ Important Distinction

To differentiate from **financial capital** (bank deposits, stocks, bonds):

This type of capital is called:

> **Fixed Capital**

🏗️ Types of Capital

Type	Meaning	Economic Role
Physical Capital	Tangible assets used in production	Increases productivity & output efficiency
Human Capital	Skills, education, knowledge, experience	Improves labor productivity
Social Capital	Networks, institutions, relationships	Builds trust & cooperation

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1️⃣ Physical Capital

Includes:

- Machines
- Buildings
- Infrastructure

📈 Effect:

- Allows firms to produce more efficiently
- Increases total output

2️⃣ Human Capital

Refers to:

- Education
- Skills
- Experience
- Training

📈 Effect:

- Better productivity
- Higher economic contribution

3️⃣ Social Capital

Includes:

- Trust
- Business networks
- Institutional strength

📈 Effect:

- Better cooperation
- Improved economic efficiency

📈 4.10 Investment

📌 What is Investment?

In economics:

> Investment = Spending on assets that increase future production

It is not buying stocks —

It is buying **productive assets**.

🏗️ Types of Investment

1️⃣ Gross Investment

Includes:

- All finished goods produced but not consumed
- New machines
- Buildings
- Equipment

🔹 Gross Fixed Investment

Specifically includes:

- Machinery
- Factories
- Infrastructure

2️⃣ Net Investment

$$\text{Net Investment} = \text{Gross Investment} - \text{Depreciation}$$

🔻 What is Depreciation?

> Wear and tear of capital over time

Examples:

- Machines breaking down
- Buildings aging

🎯 Why Net Investment Matters

Net Investment represents:

- Real increase in capital stock
- Actual economic growth

If:

- Gross Investment = Depreciation

  → Capital stock stays the same

If:

- Gross Investment > Depreciation

  → Economy grows

👷 5. The Labor Market

5.1 📌 Introduction

Labor is a key factor of production.

The total labor used affects:

- GDP
- Employment levels

A major concern in labor markets:

> **Unemployment**

Because it affects:

- Income
- Stability
- Growth

5.2 📉 Types of Unemployment

Unemployment is classified into **four types**:

1️⃣ Frictional Unemployment

Temporary unemployment while:

- Searching for jobs
- Switching careers
- Relocating

📌 Example:

- A recent college graduate looking for their first job

2️⃣ Structural Unemployment

Occurs when:

- Worker skills do not match job requirements
- Technology replaces certain jobs

📌 Example:

- Automation replacing factory workers

3️⃣ Cyclical Unemployment

Linked to:

- Business cycle

Increases during:

- Recessions

📌 Example:

- Construction worker losing job during economic downturn

4️⃣ Classical Unemployment

Occurs when:

- Wages are set too high
- Firms cannot afford to hire

📌 Example:

- High minimum wage discouraging hiring

5.3 🎯 Full Employment

Full employment does **NOT** mean zero unemployment.

It means:

> The economy is at its **natural rate of unemployment**

Includes:

- Frictional unemployment
- Structural unemployment

Excludes:

- Cyclical unemployment

📊 Economic Phases

Phase	Cyclical Unemployment
Boom	Low
Recession	High

5.4 💵 Wages

5.4.1 Nominal Wages

Nominal wage =

> The actual amount of money earned

Example:

- $20 per hour

⚠️ Does NOT account for inflation

5.4.2 Wages vs Income

Term	Meaning
Wages	Earnings per hour or job
Income	Total earnings from all sources

Income includes:

- Wages
- Investments
- Property earnings

5.4.3 Real Wages

Real wage =

> Wage adjusted for inflation

Shows:

- Actual purchasing power
- How many goods/services can be bought

💰 6. Money and Banks

6.1 💵 What is Money?

Money is:

> Anything widely accepted as payment for goods and services

🏦 Types of Money

Type	Description
Currency & Coins	Physical cash
Bank Deposits	Digital money in accounts
Money vs Wealth	Wealth includes assets; money is spendable cash

🔎 Money vs Wealth

- Wealth = Real estate, stocks, assets
- Money = Cash + deposits available for spending

🏛️ 6.2 Central Banks and Their Role

A **central bank** controls the country's monetary system.

For example:

- Reserve Bank of India (India)
- Federal Reserve (USA)

🎯 Main Functions

Function	Role
Monetary Policy	Controls interest rates & money supply
Regulating Banks	Ensures financial stability
Managing Foreign Reserves	Stabilizes exchange rate
Lender of Last Resort	Emergency loans to banks

📊 7. Interest Rates

7.1 📌 What is Interest?

Interest is:

> The cost of borrowing money

Usually expressed as:

- Percentage of loan amount

📈 Types of Interest Rates

Type	Meaning
Fixed Interest Rate	Remains constant
Floating Interest Rate	Changes with market conditions

📌 Example Calculation

If:

- Loan = $10,000
- Interest rate = 10%

Interest per year:

$10,000 \times 0.10 = 1,000$

So borrower pays:

- $1,000 per year in interest

📘 MACROECONOMIC MODEL

🧠 INTRODUCTION

🔹 What Are Macroeconomic Models?

Macroeconomic models are **analytical tools** used to understand how different factors influence the economy.

They help explain and predict how key economic variables interact when there are changes such as:

- Government spending increases
- Central bank interest rate adjustments
- Changes in foreign demand for domestic goods

These models allow economists to study cause-and-effect relationships in the economy.

📊 Policy Analysis

Macroeconomic models are widely used for policy evaluation.

Examples:

- What happens to unemployment if the central bank lowers the target interest rate?
- What are the effects of increased government spending on economic growth?

They help governments and central banks anticipate policy outcomes before implementation.

🔎 Economic Observations

Models also help analyze real-world economic patterns such as:

- Cyclical fluctuations in economic growth
- The relationship between unemployment and inflation (e.g., Phillips Curve)
- The impact of interest rates on foreign exchange rates

⚖️ The Complexity of Macroeconomics

Macroeconomics is not an exact science. It involves uncertainty and interpretation.

1️⃣ No Exact Science

Unlike physics:

- There are no universal laws
- Relationships between variables are debated
- Outcomes depend on assumptions

2️⃣ Multiple Models, Multiple Predictions

Different models can produce different results.

Example:

- One model may predict that reducing interest rates decreases unemployment.
- Another may predict little or no effect.

Both conclusions depend on the structure and assumptions of the model used.

3️⃣ Model vs Reality

Statements like:

> "An increase in X leads to an increase in Y"

Are properties of a **model**, not necessarily the real world.

Observed economic events often have multiple possible explanations.

❓ Why Do We Need Macroeconomic Models?

1️⃣ To Understand the Economy

Models explain relationships between:

- GDP
- Inflation
- Unemployment

Example:

- Increased consumer spending → Higher GDP → Lower unemployment

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2️⃣ Policy Analysis

Models predict outcomes of:

- Tax cuts
- Government spending increases

Example:

- Reducing taxes may increase household spending and economic growth.

3️⃣ Decision-Making

Central banks rely on models to decide:

- Interest rates
- Money supply changes

Example:

- The Federal Reserve may lower interest rates to reduce unemployment.

4️⃣ Simplify Complexity

The real-world economy is highly complex.

Models focus on:

- Key variables
- Essential relationships

Example:

- A trade model may only focus on exports, imports, and exchange rates.

5️⃣ Predict Economic Outcomes

Models help forecast effects of:

- Recessions
- Oil price shocks

Example:

- An oil price increase can be modeled to predict its impact on inflation and spending.

🏛️ The Neo-Classical Synthesis

A widely accepted framework in introductory macroeconomics.

It combines two major economic theories:

1️⃣ Classical Model

Focus:

- Long-run economy

Assumptions:

- Markets are efficient
- Prices and wages adjust freely
- Full employment is the long-run norm

2️⃣ Keynesian Model

Focus:

- Short-run dynamics

Key Idea:

- Prices and wages may be sticky
- Unemployment or inflation can occur during fluctuations

🔄 The Neo-Classical Synthesis

Combines both approaches:

- Keynesian model explains short-term fluctuations (recessions, inflationary periods)
- Classical model explains long-term equilibrium

Short run → Keynesian

Long run → Classical

🔄 Why Are There Variations in Models?

1️⃣ Different Assumptions

Models differ in assumptions about:

- Labor markets
- Capital mobility
- Price flexibility

2️⃣ Different Economic Goals

Some models focus on:

- Economic growth

Others focus on:

- Inflation control
- Unemployment reduction

3️⃣ Empirical Limitations

Unlike natural sciences:

- Economists cannot conduct controlled experiments
- Observations are influenced by many external factors

📌 COMMON ASSUMPTIONS OF MACROECONOMIC MODELS (FUMICE)

Macroeconomic models simplify reality using structured assumptions.

These assumptions help isolate relationships between variables.

1️⃣ Unemployment and Hours Worked

Assumption

There is a **negative relationship** between:

- Unemployment
- Total hours worked

If hours worked increase → Unemployment decreases.

Why?

More people working means fewer unemployed individuals.

Reality Check

This may not always hold.

Example:

- Increase in immigration → More labor force
- If jobs do not increase proportionally → Unemployment may rise

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Example

If a factory hires more workers:

- Hours worked increase
- Unemployment decreases

But if many immigrants enter the labor market:

- Hours worked may increase
- Unemployment may also increase

2️⃣ Central Bank Controls Money Supply

Assumption

The central bank fully controls:

- Monetary base
- Money supply (assuming constant money multiplier)

Why?

The monetary base (cash + reserves) is controlled by the central bank.

Example:

- If the Reserve Bank of India increases the monetary base, money supply expands proportionally (if multiplier constant).

3️⃣ Monetary Policy = Change in Money Supply

Assumption

Changing the target interest rate is equivalent to changing the money supply.

Why?

There is a negative relationship between:

- Interest rate
- Money supply

Lower interest rate → Higher money supply.

Example

If the central bank lowers interest rates:

- Borrowing becomes cheaper
- Spending and investment increase
- Similar effect as increasing money supply

4️⃣ One Interest Rate Assumption

Assumption

Models assume a single representative interest rate.

Why?

Different interest rates move in similar directions.

Example:

If the one-year government bond rate falls:

- Mortgage rates likely fall
- Corporate bond rates likely fall

Simplifying to one rate does not change directional predictions.

5️⃣ Exchange Rate Assumptions

Assumption

Exchange rate is flexible and determined by domestic vs foreign price levels.

If:

- Domestic prices ↑ 10%
- Foreign prices unchanged

Then:

- Domestic currency depreciates by 10%

Why?

Based on **Purchasing Power Parity (PPP)**.

Trade Implication

Exports and imports remain unaffected because depreciation offsets price increase.

Example

Domestic goods become 10% more expensive.

But currency depreciates 10%.

Result:

- Foreign buyers see no price change.
- Exports remain stable.

6️⃣ Capital Flows Assumption

Assumption

If domestic interest rates rise relative to foreign rates:

- Capital flows into the country
- Interest rates eventually stabilize

Why?

Investors seek higher returns.

Higher interest rate → More foreign investment → More funds supply → Rate falls back toward equilibrium.

Example

Domestic rate = 5%

Foreign rate = 2%

Investors shift capital domestically to earn 5%.

This inflow:

- Increases supply of funds
- Pushes interest rates downward
- Restores balance

📊 THE MACROECONOMIC VARIABLES

Macroeconomic models use a set of core variables to describe how the economy functions.

Below is a structured summary of the key variables.

🔑 Key Variables in Macroeconomic Models

Symbol	Variable	Meaning
Y	Real GDP	Total output of goods and services adjusted for inflation
U	Unemployment	% of labor force not working but actively seeking jobs
L	Hours Worked	Total number of hours worked in a given period
P	Price Level	Average price of goods and services in the economy
M	Money Supply	Total money in circulation
R	Nominal Interest Rate	Interest rate not adjusted for inflation

Symbol	Variable	Meaning
r	Real Interest Rate	Nominal interest rate minus inflation
E	Exchange Rate	Value of domestic currency relative to foreign currencies
NX	Net Exports	Exports – Imports
π	Inflation	Rate at which prices are rising

📈 Example: Analyzing Macroeconomic Changes

Scenario: Central Bank Lowers Interest Rates

1️⃣ Monetary Policy Effect

Lower interest rates → Cheaper borrowing.

Why?

- Businesses borrow at lower cost → Invest more.
- Consumers find loans (mortgages, car loans) cheaper → Spend more.

2️⃣ Impact on Key Variables

a🔳 Y (Real GDP)

What happens?

→ Real GDP increases.

Why?

- Higher business investment.
- Higher consumer spending.
- Increased aggregate demand → Higher output.

---

b🟥 U (Unemployment)

What happens?

→ Unemployment decreases.

Why?

- Higher demand → Firms hire more workers.
- More production → More employment.

---

c🟥 E (Exchange Rate)

What happens?

→ Domestic currency depreciates.

Why?

- Lower interest rates → Lower returns for foreign investors.
- Capital flows out → Demand for currency falls.
- Currency weakens.

Effect:

- Exports become cheaper.
- Exports increase.

---

d🟥 P (Price Level)

What happens?

→ Price level rises (Inflation increases).

Why?

- Higher spending increases demand.
- If supply cannot expand quickly → Prices increase.

🏦 Real-Life Example

After a recession:

- Central bank cuts interest rates.
- Firms expand production.
- Consumers borrow more.
- GDP rises.
- Employment rises.
- Inflation increases moderately.

📉 SUPPLY AND DEMAND

Microeconomic Distinction

1️⃣ Demand

Quantity consumers want to buy at different prices.

Example: Lower apple prices → Higher demand.

2️⃣ Supply

Quantity producers are willing to sell at different prices.

Example: Higher apple prices → More apples supplied.

3️⃣ Observed Quantity

Actual quantity bought and sold (equilibrium).

Example: If 500 apples are demanded and supplied → Observed quantity = 500.

Functions vs Variables

- **Demand & Supply** → Functions (depend on price, income, etc.)
- **Observed Quantity** → Variable (single measurable value)

Graphical Explanation

- Demand curve → Downward sloping.
- Supply curve → Upward sloping.
- Intersection → Equilibrium price & quantity.

🌍 Macroeconomic Supply and Demand

In macroeconomics, supply and demand apply to many variables:

Total Hours Worked (L)

- **LS** → Labor Supply (hours people want to work)
- **LD** → Labor Demand (hours firms want)

Other Variables

Variable	Supply & Demand Concept
Y (GDP)	Supply & demand for goods
K (Capital)	Investment demand & capital supply
M (Money)	Money supply & money demand
C (Consumption)	Household demand
I (Investment)	Firms' demand for capital
G (Gov Spending)	Government demand
X (Exports)	Foreign demand
Im (Imports)	Domestic demand for foreign goods

Subscript Notation

- **S** → Supply (LS, MS)
- **D** → Demand (LD, MD)

- No subscript → Observed value (L, M)

Example: Labor Market

- LD = 1,000 hours
- LS = 800 hours
- Observed L = 800 hours (lower of supply and demand)

🌱 GROWTH MODEL (Classical Growth Theory)

Developed by **Thomas Robert Malthus** (late 1700s).

Core Mechanism

1️⃣ Technological Development

→ Raises productivity.

2️⃣ Higher Productivity

→ Increases GDP per capita.

3️⃣ Higher Living Standards

→ Population grows.

4️⃣ More People, Same Resources

→ Productivity falls.

5️⃣ GDP per Capita Returns to Survival Level

Capital Destruction

If war destroys capital:

- Productivity falls.
- Population declines.

- Fewer people share resources.
- GDP per capita returns to survival level.

Main Conclusion

Population growth cancels out technological gains.

GDP per capita always returns to survival level.

This pessimistic view led to economics being called

“the dismal science.”

Why It Was Wrong

In the 1800s:

- Industrial Revolution boosted productivity massively.
- Institutions improved.
- Education improved.
- Population growth eventually slowed.

Technological progress outpaced population growth.

🚀 NEO-CLASSICAL GROWTH MODEL (Solow Model)

Developed by **Robert Solow** (1960s).

Also called:

- Solow Growth Model
- Exogenous Growth Model

Core Idea

Permanent GDP per capita growth comes only from:

> Technological Progress

Key Differences from Classical Model

Classical Model	Neo-Classical Model
Population depends on income	Population growth is exogenous
GDP returns to survival level	GDP per capita can grow permanently
Tech gains offset by population	Continuous tech progress sustains growth

Role of Savings

- Higher savings → More capital.
- Temporary GDP per capita increase.
- Eventually reaches new steady state.
- Long-run growth determined by technology.

Role of Government

- Can affect growth temporarily.
- Cannot permanently change growth rate.
- Long-run growth depends on technological progress.

What Drives Long-Run Growth?

Only continuous technological improvement.

Not:

- Higher savings
- Higher government spending
- Temporary capital accumulation

Example

1️⃣ Workers use tools.

2️⃣ Technology improves tools.

3️⃣ Output increases.

4️⃣ Population grows.

5️⃣ Continued technological progress → Sustained growth.

🚀 ENDOGENOUS GROWTH THEORY

Developed in the 1980s by **Paul Romer** and others.

Unlike the Solow (neo-classical) model:

- Technological progress is **explained inside the model**
- It is **endogenous**, not exogenous

🔑 Core Ideas (TCP-PEGS)

1️⃣ Technology Comes From Inside the Economy

Countries grow because people:

- Study
- Invent
- Experiment
- Get better through practice

➡️ Tech isn't magic from outside. People create it.

Simple example:

More research money → new apps, new medicine, smarter AI.

2️⃣ More Capital Can Keep Helping (Not Just At First)

In this theory:

➡️ More machines, knowledge, and tools can **keep boosting growth**, not just temporarily.

Simple example:

Learning + computers + software → productivity keeps rising.

3️⃣ Technology Can Help Many People At Once

Knowledge can be shared.

➡️ One invention can help millions or billions.

Simple example:

The internet was invented once → now everyone can use it.

4️⃣ Education Is Super Important

Smarter, better-trained people → more ideas → more growth.

Simple example:

More engineers → more inventions → stronger economy.

5️⃣ Government Policies Can Change Long-Term Growth

If governments support:

- Schools
- Research
- Innovation
- Infrastructure

➡️ Growth can stay higher forever (not just temporarily).

6️⃣ Saving More Can Increase Long-Term Growth

If people save and invest more → more money goes into:

- Research
- Technology

- Education

➡️ Economy can grow faster for longer.

7️⃣ Poor Countries Can Catch Up

Because knowledge spreads.

➡️ Countries don't have to invent everything from zero — they can copy and improve.

Simple example:

A country adopts existing tech → grows faster.

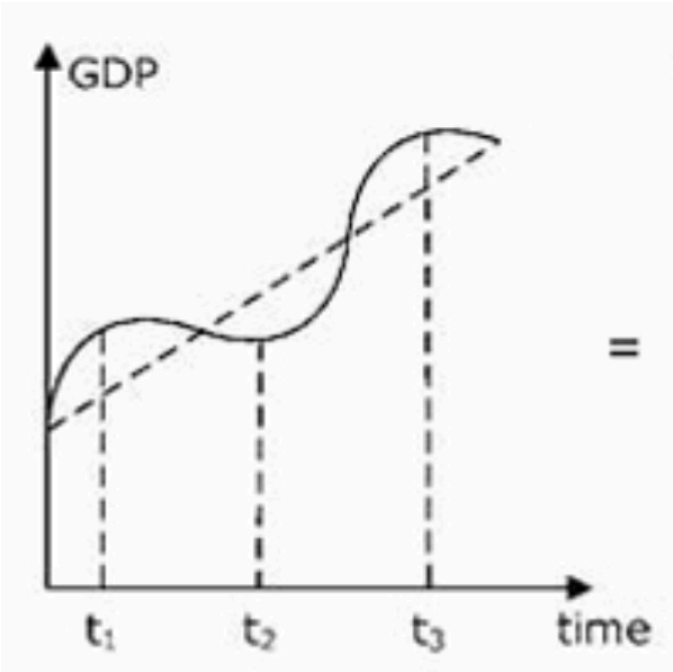
📈 Separation of Growth and Fluctuations

This is where your diagram comes in.

GDP has two components:

$$GDP = Trend + CyclicalFluctuations$$

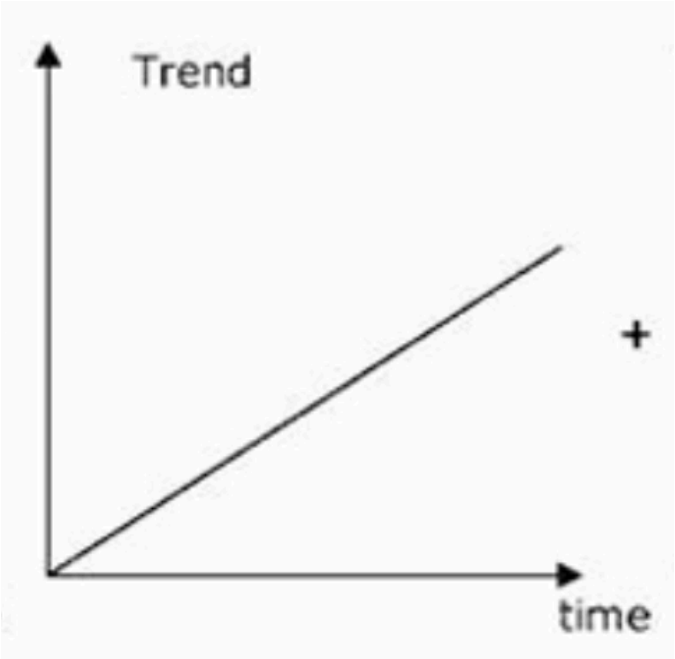
📊 Diagram 1: Combined GDP Path



Shows:

- Upward long-run trend
- Short-run ups and downs

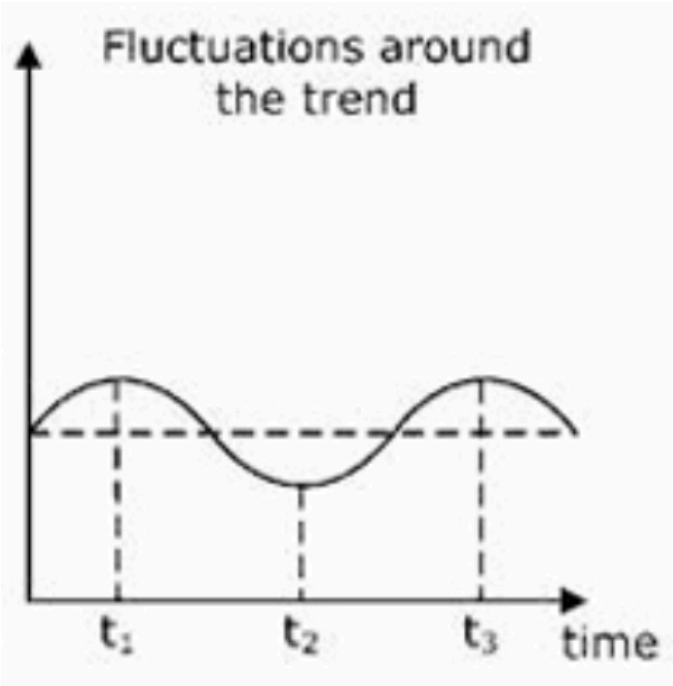
📈 Diagram 2: Trend Only



Represents:

- Long-run growth
- Driven by technology
- Focus of growth theory

📉 Diagram 3: Cyclical Fluctuations



Represents:

- Booms and recessions
- Short-run deviations from trend

🎯 Why Separate Them?

Growth:

- Long-term
- Driven by technology
- Human capital
- Innovation

Fluctuations:

- Short-term
- Demand shocks
- Financial crises

- Policy shocks

Example:

- 2008 crisis → fluctuation
- 50-year productivity growth → trend

🏛️ THE CLASSICAL MODEL (Simple Explanation)

Popular before the Great Depression.

🔑 Core Assumptions

1️⃣ Prices and Wages Adjust Freely

- Demand rises → wages rise
- Demand falls → wages fall

Markets clear automatically.

2️⃣ Full Employment Always

Economy always operates at:

$$PotentialOutput(Y^*)$$

Everyone willing to work gets a job.

3️⃣ Self-Correcting Economy

If recession happens:

- Wages fall

- Firms hire more
- Economy returns to full employment

No government needed.

📊 Classical Model Diagram

Long-Run Aggregate Supply (Vertical)

```
PriceLevel(P)
     |
     |        LRAS
     |          |
     |          |
     |          |
     |          |
     |__________|________Output(Y)
                Y*
```

- Output fixed at potential (Y*)
- Only prices adjust
- Economy always at full employment

Highway Analogy

Economy = highway

If traffic slows:

- Natural adjustment restores speed

No traffic police needed.

❌ Limitation of Classical Model

Great Depression showed:

- Wages didn't fall fast enough
- Unemployment remained high
- Economy didn't self-correct quickly

This led to Keynesian revolution.

📘 KEYNESIAN CROSS MODEL

Introduced by **John Maynard Keynes** in *The General Theory of Employment, Interest, and Money* (1936).

It emerged during the **Great Depression** to challenge core Classical assumptions.

🧠 Why It Was Revolutionary

The Classical Model assumed:

- Markets self-correct
- Full employment always
- Supply creates its own demand

Keynes disagreed.

He argued:

> Demand shortages can persist.
>
> The economy can remain stuck below full employment.

🔍 Key Differences: Classical vs Keynesian

Issue	Classical Model	Keynesian Model
Say's Law	Supply creates its own demand	Demand can be insufficient
Money	Money affects only prices	Spending & investment matter more

Issue	Classical Model	Keynesian Model
Wages	Flexible	Sticky (slow to adjust)
Employment	Always full	Can have persistent unemployment
Government Role	Minimal	Active intervention needed

1️⃣ Say's Law Does Not Apply

Classical:

> Everything produced will be purchased.

Keynes:

- Demand shortages can exist.
- Firms may not sell all output.
- Leads to unemployment & unused capacity.

2️⃣ Quantity Theory of Money Does Not Apply

Classical:

Money supply ↑ → Prices ↑

Keynes:

- Spending behavior matters more.
- Investment & consumption drive output.

3️⃣ Nominal Wages (W) Are Exogenous

In the Keynesian model:

- Wages are predetermined (exogenous).
- They do not adjust quickly.

Why?

- Labor contracts
- Minimum wage laws
- Worker resistance

For simplicity in Cross Model:

W=constant

🏗️ Stages of Keynesian Model Development

The Keynesian framework evolves step-by-step.

1️⃣ Stage 1: Keynesian Cross

Assumptions:

- W constant
- P constant
- R constant

Focus:

> Aggregate Demand determines Output.

2️⃣ Stage 2: IS–LM Model

Assumptions:

- W, P constant
- R endogenous

Focus:

- Goods market (IS)
- Money market (LM)

- Determines Y and R

3️⃣ Stage 3: AS–AD Model

Assumptions:

- W constant
- P & R endogenous

Focus:

- Price level and output interaction

4️⃣ Stage 4: Full Keynesian Model

Assumptions:

- W exogenous (but adjustable over time)
- P & R endogenous

Focus:

- More realistic long-run dynamics

📊 Stage 1: Keynesian Cross Diagram (not required)

Axes:

- Horizontal → Output / Income (Y)
- Vertical → Planned Expenditure (E)

```
Planned Expenditure (E)
        |
        |45° line
        |       /
        |      /
```

```
        |      /
        |     /
        |    /
        |   /       AE
        |  /       /
        |/_______/____________ Y
             Y*
```

Explanation:

- 45° line → Points where Output = Expenditure
- AE curve → Aggregate Expenditure
- Intersection → Equilibrium output

If AE shifts upward → GDP increases.

📘 Stage 2: IS–LM MODEL

Developed by **John Hicks (1937)**.

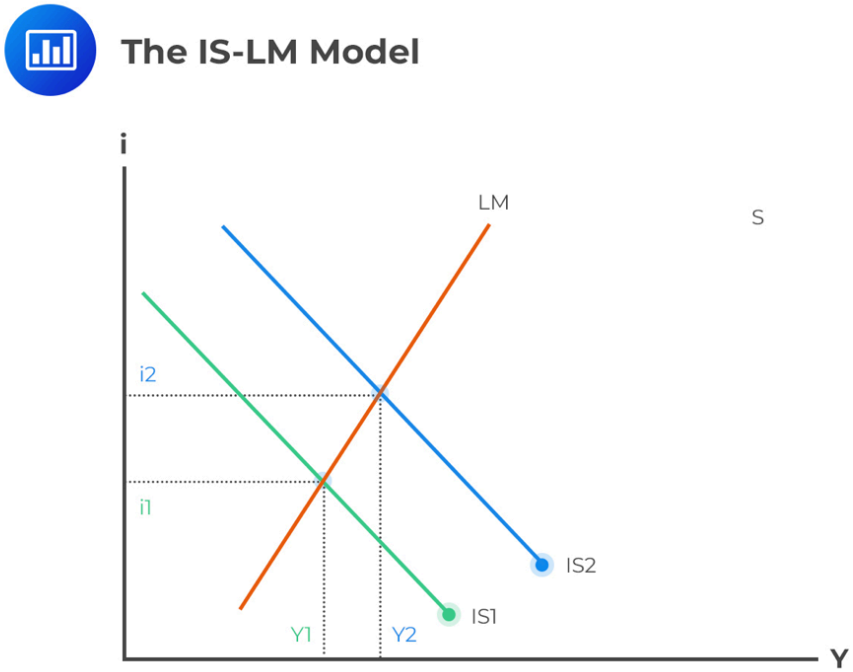
Shows interaction between:

- Goods market (IS)
- Money market (LM)

📊 IS–LM Diagram

Axes:

- Horizontal → GDP (Y)
- Vertical → Interest Rate (R)



🔹 IS Curve

- Downward sloping
- Lower interest rate → Higher investment → Higher output

🔹 LM Curve

- Upward sloping
- Higher income → Higher money demand → Higher interest rate

🔹 Equilibrium

Intersection = Short-run equilibrium (Y∗, R∗)

💰 Expansionary Monetary Policy (IS–LM)

Money supply increases → LM shifts right.

Result:

- Interest rate ↓
- Output ↑

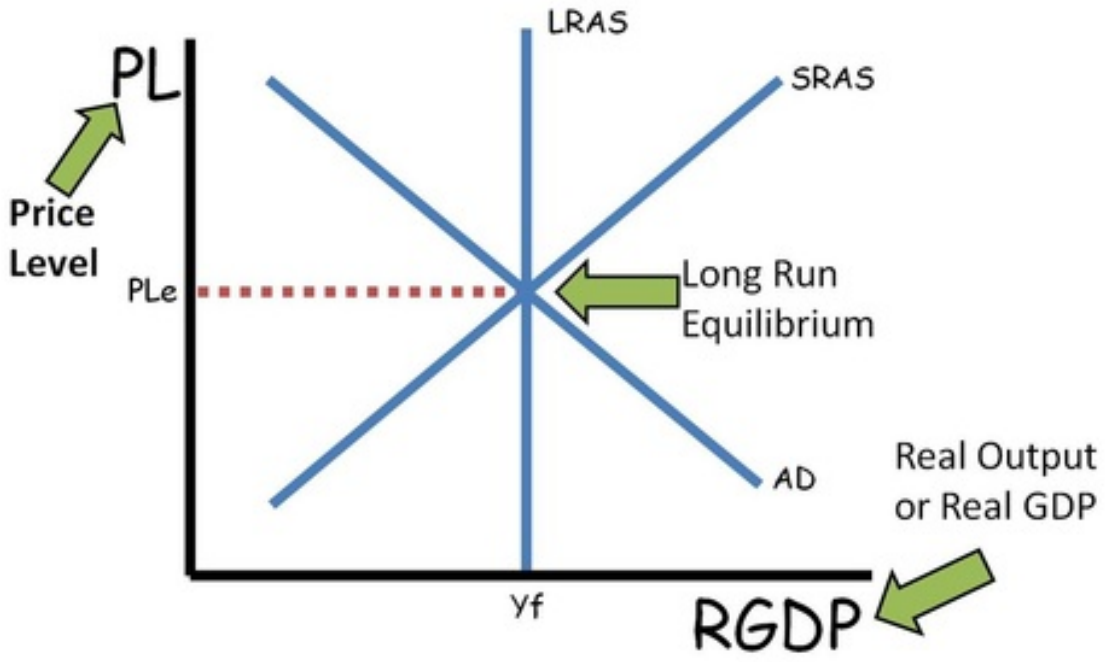
📘 Stage 3: AS–AD MODEL

Most important macro diagram.

📊 Basic AS–AD Diagram

Axes:

- Horizontal → Real GDP (Y)
- Vertical → Price Level (P)



Intersection = Equilibrium (Y∗, P∗)

📈 Expansionary Monetary Policy (AS–AD)

AD shifts right.

```
Price Level
      |
      |        AS
      |       /
      |      /
 P2   |-----/------ AD2
 P1   |----/----- AD1
      |   /
      |  /
      | /
      |/_________________ GDP
            Y1   Y2
```

Result:

- GDP ↑
- Price level ↑

📉 Contractionary Monetary Policy

AD shifts left.

```
Price Level
      |
      |        AS
      |       /
      |      /
 P1   |-----/------ AD1
 P2   |----/----- AD2
      |   /
      |  /
      | /
```

```
        |/_________________ GDP
              Y2    Y1
```

Result:

- GDP ↓
- Inflation ↓

📊 Fiscal Policy (From Your Diagram)

Fiscal policy also shifts AD.

Expansionary Fiscal Policy

(G ↑ or T ↓)

AD → Right

GDP ↑

Inflation ↑

Contractionary Fiscal Policy

(G ↓ or T ↑)

AD → Left

GDP ↓

Inflation ↓

🏛️ Short Run vs Long Run (AS–AD)

Short Run:

- SRAS upward sloping
- Output can deviate from potential

Long Run:

- LRAS vertical at potential output (Y*)

```
Price Level
      |
      |         LRAS
      |          |
      |          |
      |          |
      |          |
      |__________|_________ GDP
                 Y*
```

Long-run output determined by:

- Technology
- Capital
- Labor

🎯 Summary: What Each Model Explains

Model	Explains
Keynesian Cross	Output from demand (fixed prices)
IS–LM	Interest rate & output interaction
AS–AD	Price level + output
Classical	Long-run full employment
Growth Models	Long-term GDP trend